

PRESENTATION ON

INVENTORY MANAGEMENT & PURCHASE OPTIMIZATION

Analysis for Rajesh Surgical

PRESENTED BY:

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ABOUT BUSINESS



- ***Rajesh Surgical is a 30-year-old medical wholesale business based in Kaji Ki Sarai, Kothiwalan, Moradabad (U.P.).***
- ***Founded by Mr. Rajesh Sharma; currently managed by Ankit Pandey & Ishita for the last 9–10 years.***
- ***Handles 1500+ products across surgical consumables, disposables, and medical devices.***
- ***Acts as a key distributor connecting manufacturers with hospitals, clinics, and local retailers.***
- ***Although records are digital, purchasing & inventory decisions rely mostly on experience, not data.***
- ***This project converts the available data into actionable insights for forecasting, inventory control, and cost optimization.***

ABOUT PROJECT

AIM/ OBJECTIVE

Develop a data-driven system to improve forecasting, inventory control, and procurement decisions for Rajesh Surgical.

Problem Statement 1

INEFFECTIVE INVENTORY PLANNING

Rajesh Surgical lacks a data-driven method to understand demand and plan stock levels. This leads to overstocking slow movers and understocking high-demand items, causing capital lock-up and missed sales.

Problem Statement 2

INEFFICIENT COST MANAGEMENT

Purchasing decisions do not account for price fluctuations. Without analyzing rate trends, the company misses low-price opportunities and spends more on procurement.

DETAILS ABOUT DATA

ABOUT DATA:

- Daily purchase records maintained digitally by Rajesh Surgical
- Time period analysed: Apr 2024 – Mar 2025 (cost & rate analysis)
- Historical extension: Apr 2022 – Mar 2025 (for forecasting)
- **Columns:** Date, Item Description, Quantity, Rate, Amount
- **Product Range:** 1500+ surgical consumables & medical devices

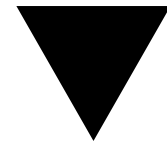
TOOLS USED:

- Excel
- Python

RAJESH SURGICAL							
KAJI KI SARAI,KOTHIWALAN, MORADABAD							
Phone : 9219537999,7060280809 E-Mail : ANKITPANDAY139@GMAIL.COM							
Licence No. : UP2120B000921,UP2121B000917							
GSTIN : 09ASXPS6634D1Z1 TIN No. : 09359001820 Food Lic.No. : 22722722000007							
PURCHASE SUMMARY							
01-04-2024 - 31-03-2025							
Date	Bill	Item Description	Batch	Qty.	Free	Rate	Amount
01-04-2024	000001	DISPO. 3PLY	ELAS		35	85.7160	3000.06
01-04-2024	000001	DISPO. 3PLY	ELAS		240	75.0000	18000.00
01-04-2024	000001	SHOE COVER	PLAST		60	160.0000	9600.00
01-04-2024	000001	SAMA GUAZE	SMDGS3801		1800	11.1625	20092.50
01-04-2024	000001	ARC 4*3.6M	23072502		260	95.0000	24700.00
01-04-2024	000001	ARC 3*3.6M	23072502		201	85.0000	17085.00
01-04-2024	000001	CAP BUFFERANT			160	52.0000	8320.00
01-04-2024	000001	CISLON PLUS	100M AL2403050		200	9.0000	1800.00
01-04-2024	000001	CISLON PLUS	1000 AL2403046		45	44.0000	1980.00
01-04-2024	000001	HANDLOOM GAUGE	S 054		500	14.2880	7144.00
01-04-2024	119	AVIL INJ. 2ML	2123114		1800	4.4019	7923.48
	01-04-2024	2425/0002 BIOS DNS CP	500M CDS24029		1920	12.0920	23216.64
	01-04-2024	2425/0002 BIOS NS FFS	100M CNS24012		2000	6.6500	13300.00
	01-04-2024	2425/0002 BIOS NS CP	500ML CNB24070		1200	12.0920	14510.40
	01-04-2024	2425/0002 BIOS RL CP	500ML CRL24061		960	12.4600	11961.60
	01-04-2024	2425/0002 BIOS OFLO MONO	F OFF23040		2000	9.1000	18200.00
16		01-04-2024			13381		200833.68

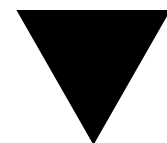
APPROACH

Data Collection



Data Processing

- Merged all raw files, and standardized column names.
- Corrected misaligned rows, removed duplicates, and cleaned invalid entries.
- Standardized item descriptions



Data Selection

Focus on Quantity, Rate, Amount, Date, and Item Description

ANALYSIS

Problem Statement 1

- Performed ABC classification
- Conducted VED analysis on A-category items to derive AV (A-Vital) items requiring strict control.
- Classified these AV items into Regular and Intermittent demand groups using CV and Zero-Months %.
- Applied Holt-Winters & Croston's Method for forecasting.

Problem Statement 2

- Identified volatile-rate items among AV items.
- Price trend and variability analysis
- Determined optimal bulk purchase month

RESULTS AND FINDINGS

ABC ANALYSIS:

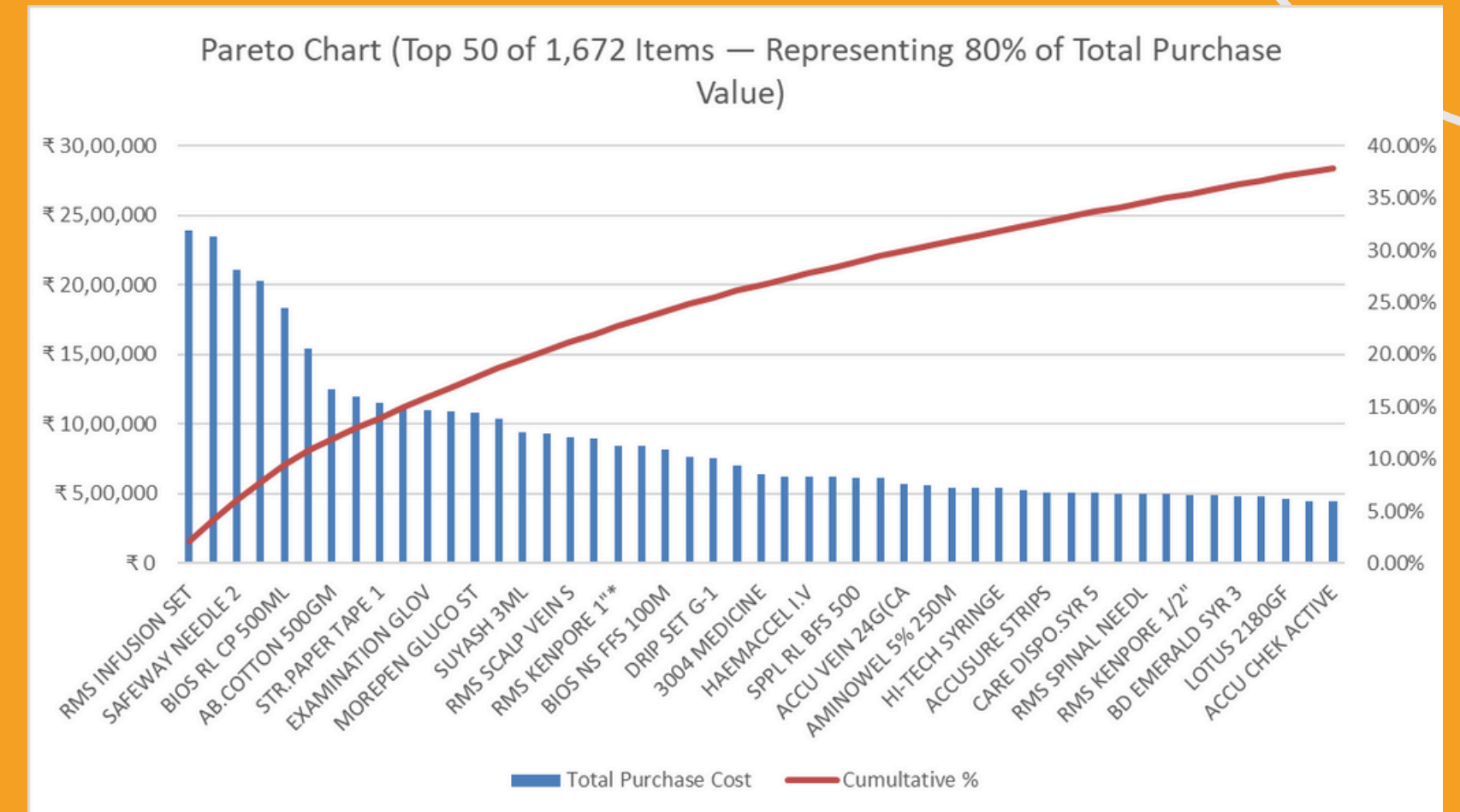
- Applied ABC analysis on 1,671 unique items to rank products by annual purchase value.
- Identified high-value items (A-category) contributing most to total expenditure.

VED CLASSIFICATION:

- Performed VED analysis on A-category items to assess their criticality.
- Classified items as Vital (V), Essential (E), and Desirable (D).

OUTCOME:

- Identified 58 (initially 61) AV (A-Vital) items, representing both high-value and high-criticality products.
- These 58 items were selected for detailed forecasting and cost analysis.



Category	No. of Items	% of Total Items	Total Purchase	Purchase %
A	327	20%	₹ 9,08,08,953	80%
B	448	27%	₹ 1,70,95,107	15%
C	896	54%	₹ 56,95,223	5%
Total	1671	100%	₹ 11,35,99,284	100%

Category	V	E	D	Grand Total
A	61	35	231	327
B	44	33	371	448
C	27	34	835	896
Grand Total	132	102	1437	1671

RESULTS AND FINDINGS

DEMAND CLASSIFICATION:

Used two indicators to classify AV items:

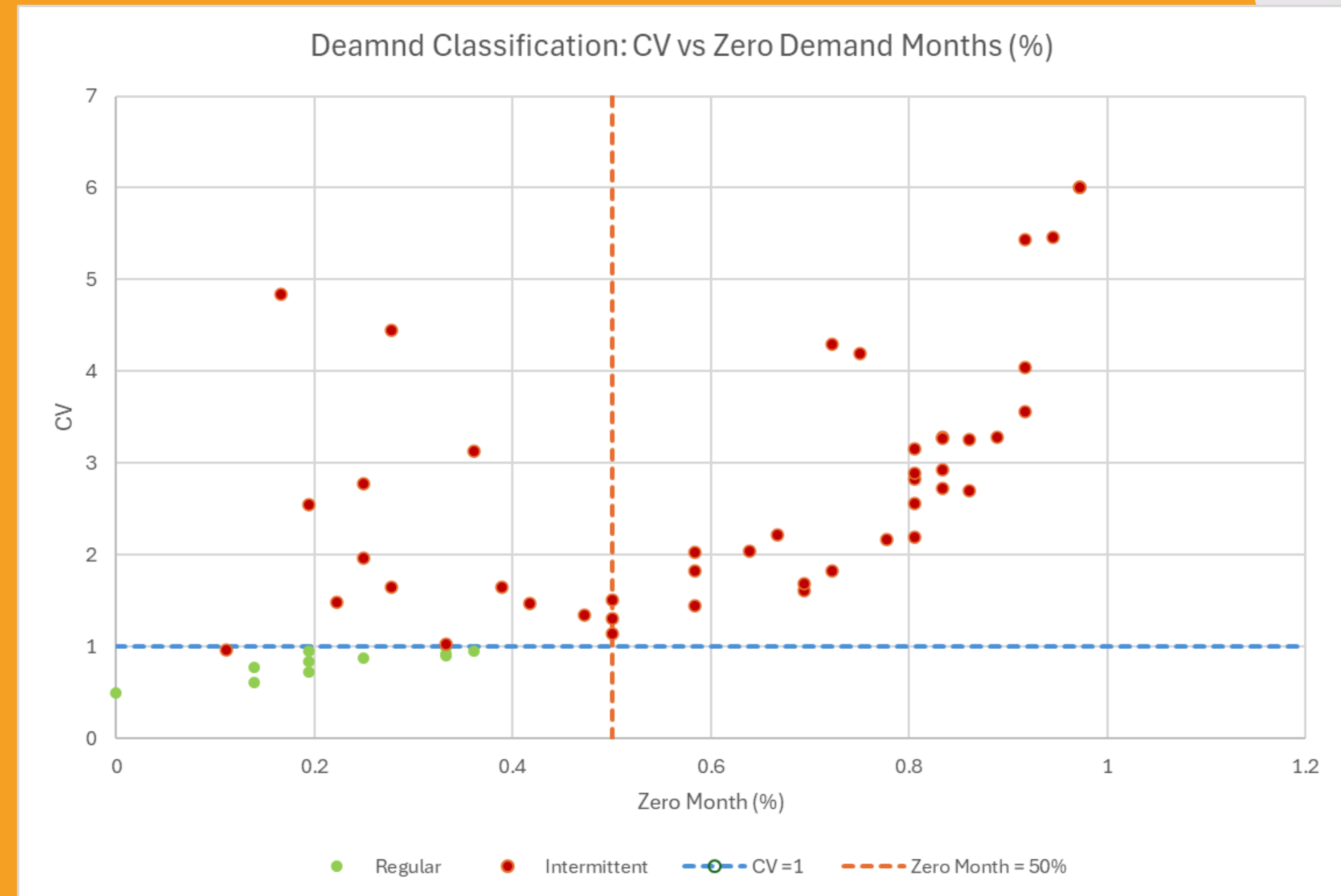
- Coefficient of Variation (CV) – measures demand variability
- Zero-Months % – months with no purchases

CLASSIFICATION CRITERIA:

- Regular Demand:
 $CV \leq 1$ and $\text{Zero-Months}\% \leq 50\%$
- Intermittent Demand:
 $CV > 1$ or $\text{Zero-Months}\% > 50\%$

OUTCOME:

- 13 Regular-demand items (stable monthly consumption)
- 45 Intermittent-demand items (irregular, tender-based or episodic purchases)



RESULTS AND FINDINGS

MODELS USED FOR FORECASTING:

Holt–Winters (Regular Demand)

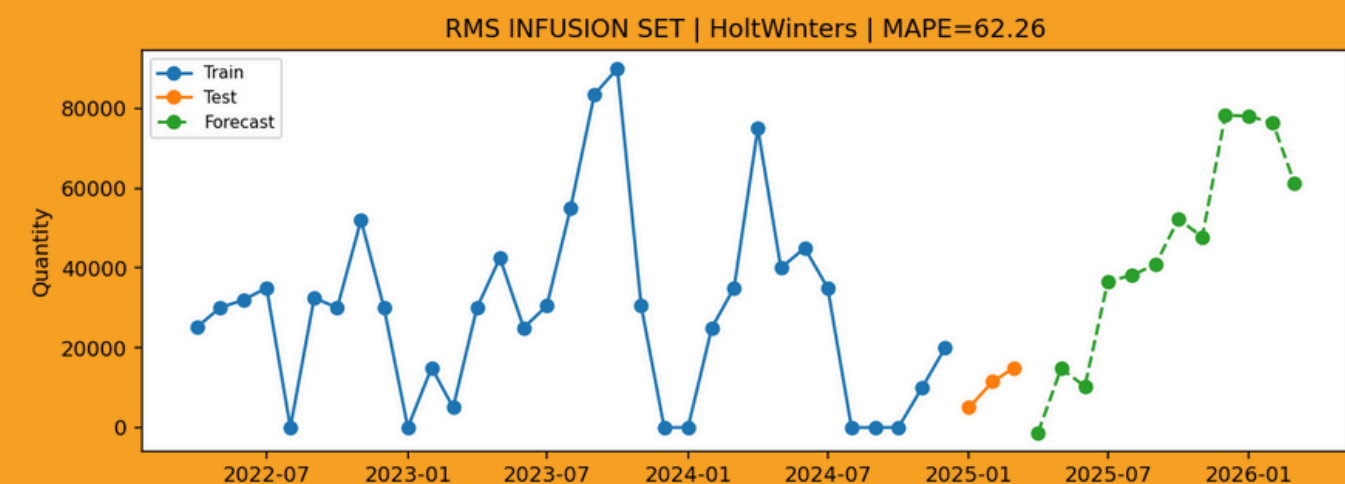
- Captures trend + seasonality for 13 stable-demand items

Croston's Method (Intermittent Demand)

- Best for irregular, zero-demand months; used for 45 items
- Post-processed to distribute forecasts across months realistically

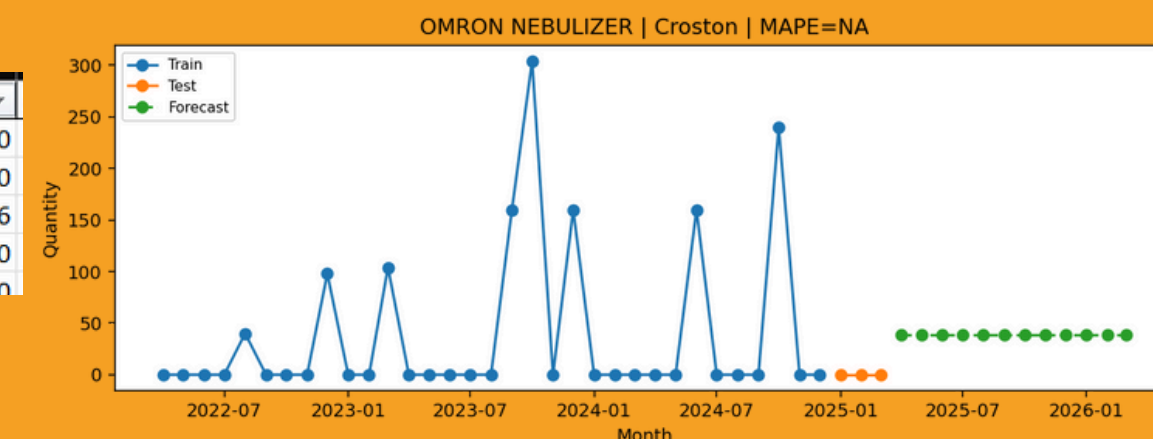
NOTE: The constant forecast value reflects Croston's behavior for intermittent items—where demand occurs in irregular bursts with many zero months. Instead of predicting monthly spikes, the model estimates the average demand per period, which helps plan replenishment only when needed rather than stocking continuously.

Product_Name	Forecast_Month	Forecast_Quantity	Avg_Rate_used	Forecast_Amount	Model	Demand_Type
INFUSION SET	2025-04-01	-1357	₹ 9.20	-₹ 12,477.13	HoltWinters	Regular
INFUSION SET	2025-05-01	14773	₹ 9.20	₹ 1,35,864.35	HoltWinters	Regular
INFUSION SET	2025-06-01	10322	₹ 9.20	₹ 94,933.31	HoltWinters	Regular
INFUSION SET	2025-07-01	36635	₹ 9.20	₹ 3,36,924.02	HoltWinters	Regular
INFUSION SET	2025-08-01	38147	₹ 9.20	₹ 3,50,835.14	HoltWinters	Regular
INFUSION SET	2025-09-01	40870	₹ 9.20	₹ 3,75,878.14	HoltWinters	Regular
INFUSION SET	2025-10-01	52233	₹ 9.20	₹ 4,80,383.13	HoltWinters	Regular



Product_Name	Forecast_Mon	Forecast_Quant	Avg_Rate_us	Forecast_Amou	Model	Demand_Ty
OMRON NEBULIZER	2025-04-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-05-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-06-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-07-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent
OMRON NEBULIZER	2025-08-01	39	₹ 1,060.00	₹ 41,374.22	Croston	Intermittent

Product_Name	Forecast_Month	Post_Forecast_Quant
OMRON NEBULIZER	2025-04-01	0
OMRON NEBULIZER	2025-05-01	0
OMRON NEBULIZER	2025-06-01	156
OMRON NEBULIZER	2025-07-01	0
OMRON NEBULIZER	2025-08-01	0

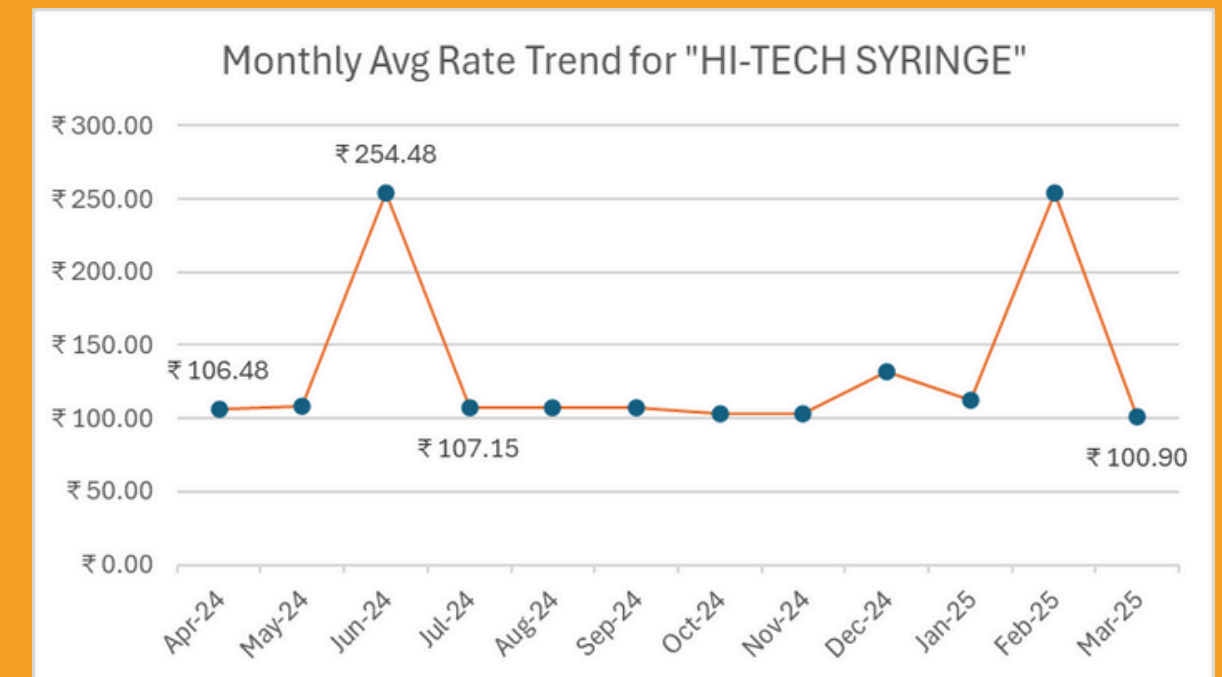


RESULTS AND FINDINGS

PRICE VARIABILITY ANALYSIS:

- For each AV item, Mean Rate, Min Rate, Max Rate, and CV of Rate were computed.
- Several items showed significant rate volatility, indicating inconsistent supplier pricing.
- Items with higher CV_Rate were tagged as priority candidates for cost optimization.

Product	Mean_Rate	Max_Rate	Min_Rate	StdDev_Rate	CV_Rate
HI-TECH SYRINGE	133.1376943	254.4766667	100.8976923	57.20820207	0.429691999
DRIP SET G-1	2.583391664	3.7	1.6965	0.935012606	0.361932191
BD EMERALD SYR 3	3.416219415	7.092105263	3	1.161087828	0.339875074
DISPOVAN NEEDLE	83.21715624	143.5	64	24.30082568	0.292017016



OPTIMAL PURCHASE TIMING & COST-SAVING POTENTIAL

- Lowest-rate month(s) identified for each volatile item.
- Potential Saving: $(\text{Mean Rate} - \text{Min Rate}) / \text{Mean Rate} \times 100$.
- Sample findings:
- HI-TECH SYRINGE: ~32% savings possible (best buying month: March).

Product_Name	Mean_Rate	Max_Rate	Min_Rate	CV_Rate	Recommended_Buy_Month	Potential_Saving_%
HI-TECH SYRINGE	133.137694	254.476667	100.897692	0.430	Mar	31.95%
EXAMINATION GLOV	108.784892	135.148148	84	0.193	Jan	29.51%
OXYGEN CONCENTRA	27827.4983	33928.57	23215	0.140	Feb-Mar	19.87%
FLAM.WHEEL CHAIR	4259.65	4812.7	3700	0.121	Nov-Mar	15.13%

RECOMMENDATIONS

- Use separate forecasting models for regular items (Holt–Winters) and intermittent items (Croston) to improve demand accuracy.
- Maintain priority stock levels for the 58 AV items identified through ABC–VED classification.
- Purchase volatile-price items during identified low-rate months to achieve 15–30% cost savings.
- Use historical rate variability to negotiate better vendor contracts and volume discounts.
- Set reorder points for intermittent items to avoid urgent, high-cost purchases.
- Allocate working capital based on demand trends, focusing more on high-consumption AV items.
- Track monthly price trends to plan bulk purchases strategically.
- Integrate forecasting and price monitoring into Excel for continuous, data-driven decision-making.
- Reduces reliance on experience-based decisions; increases transparency and predictability.
- Lowers carrying costs, avoids emergency purchases, and improves margins through smarter buying.